PART 1.

Assessment of Stevens Flood Advisory System (SFAS) service and predictions for Nor'Easter Kori

Four days before the event, between 9am and 12pm EST January 19th 2017, automated Flood Watch notices were sent to SFAS subscribers, warning them of a coming flood event. These notices were issued for a total of 35 stations (Table 1). Most notices were followed next day with a 3-day out flood watch notice.

Table 1: SFAS 4-day Flood Watches issued for Nor'Easter Kori on January 19, 2017. Watches for which a 3-day Flood Advisory was also issued are highlighted in Bold to show consistency of early forecasts. Short-term Flood Advisories issued in advance of peak heights the day of the storm, January 23, 2017, are underlined. Automated Early Flood Watches were issued for a total of 35 coastal stations, with a 30-station subset included in Automated Flood Advisory e-mails the day of the storm. URL color shows the verifying NWS local flood conditions: Above Minor (Yellow), Above Moderate Flood Level (Orange) or Above Major Flood Level (Red).

Sandy Hook NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N021

Atlantic City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N022

Cape May NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N023

Lewes DE: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N030

Ocean City Inlet MD: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N031

Great South Bay at Lindenhurst NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U204

Hudson Bay at Freeport NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U205

Raritan River at South Amboy NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U217

Raritan Bay at Keansburg NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U218

Shrewsbury River at Sea Bright NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U219

Shark River at Belmar NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U220

Barnegat Bay at Barnegat Light NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U225

Little Egg Inlet near Tuckerton NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U227

Absecon Creek Rte 30 at Absecon NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U228

Great Channel at Stone Harbor NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U235

Cohancey River at Greenwich NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U239

Watson Creek at Manasquan NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U249

FDU Bridge at Hackensack NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_1

Saw Mill Creek at Kearny NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_2

Barge Marina at Carlstadt NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_4

Kings Point NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N016

Lewisetta VA: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N047

Windmill Point VA: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N051

Hackensack River at Hackensack NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U215

Tivoli Bays South NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=HRTVBSH

Annapolis MD: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N038

Barnegat Bay at Mantoloking NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U222

Ludlam Thorofare at Sea Isle City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U233

Reynolds Channel at Point Lookout NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U206

East Rockaway Inlet at Atlantic Beach NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U207

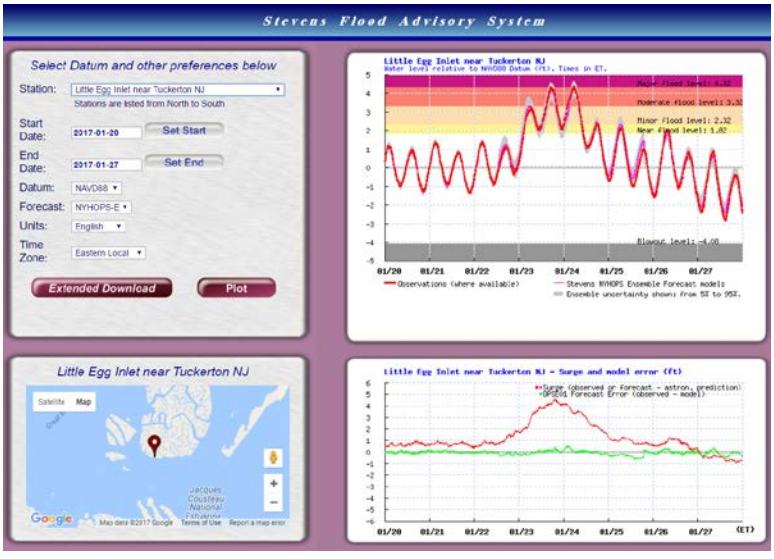
Peck Bay at Ocean City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U231

Cambridge MD: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N041

Inside Thorofare Rte 40 Atlantic City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U229

New Haven CT: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N010

East Thorofare at Ship Bottom NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U226



The local major flood level was reached at Little Egg Inlet near Tuckerton NJ at 1/23/2017 16:48 and then exceeded on the next high tide on 1/24/2017 5:30 by 0.04ft.

The automatically -mailed Flood Watch message from SFAS read:

There is a possibility of significant flooding (water levels rising above moderate flood stage) for the station(s) listed below 4 days out from now. We suggest that you use the link for each station below to check the predictions, updated every 6 hours.

SFAS Flood Advisory Notices were automatically sent to subscribers at 6am and 12pm EDT, January 23th. Stations for which these short-term Flood Advisories were sent are underlined on Table 1 at the top of the document. Flood Advisories were sent for a total of 30 stations. Importantly, and as seen on Table 1, Flood Watches had also been issued for all these 30 stations four days in advance of the storm, so ample warning existed. 5 other stations received early Flood Watches, but no Flood Advisories the day of the storm (Table 1). Thus, a total of 30 out of 76 SFAS coastal stations received a Flood Watch or Advisory. For almost all stations, the 5%-to-95% ensemble uncertainty shown in SFAS four days prior to the event enveloped the event's local observed peak water level. The 4-day-forecast cross-station average probability of exceedance of the verified January 23th event peaks was 35% (the event verified at the 65% percentile of the 96hour early prediction envelope).

An example of the e-mailed SFAS Flood Advisories and the information they contained is given below:

We expect flooding for the station(s) listed below at some point in the eight-hour period between 2017-01-23 6:00 PM and 2017-01-24 2:00 AM local time. We have indicated the approximate time when we expect the first event to occur for each station. Note that there may be higher flooding later in the eight-hour period, and we suggest that you click on the link for each station below to see complete flooding information for that station.

FDU Bridge at Hackensack NJ at approximately: 2017-01-23 7:15 PM
http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_1

Of the 30 issued Flood Advisories, 13 verified as minor floods, 15 verified as moderate floods, while 2 verified as major floods (Table 1). Three more stations listed on Table 1 were misses: Minor flooding was observed at Kings Point NY and Tivoli Bays South NY, while Moderate flooding was observed at Barnegat Bay at Mantoloking NJ. Even though an early Flood Watch was issued for these stations (Table 1), no short-term Flood Advisory was issued the day of the storm. Eight misses for which no Flood Watch or Advisory were issued (not listed on Table 1) occurred at Duck NC, at Montauk NY, at Jamaica Bay at Inwood NY, at Hudson River at South Dock at West Point NY, at Maurice River at Bivalve NJ, at New London CT, at Bridgeport CT and at Hudson River near Jersey City NJ; Minor Flood Levels were exceeded by 0.7ft, 0.6ft, 0.5ft, 0.5ft, 0.4ft, 0.3ft, 0.3ft and 0.2ft, respectively. Five more near-misses were at New York Harbor near Red Hook NY, Hudson River South of Hastings-on-Hudson NY, Hudson River below Poughkeepsie NY, Chesapeake Bay Bridge Tunnel VA and Bergen Basin at Jamaica Bay NY where levels just exceeded Minor Flood, by less than 0.1ft. A summary is found on Table 2. These 13 misses were less than 0.7ft above Minor Flood levels and did not experienced above Moderate-level flooding, which means that the early predictions were partially verified: The uncertainty envelope of the 4- and 3-days-prior predictions did not include a possibility of above moderate flooding at any of these stations, which is the condition required for SFAS to automatically issue a Flood Watch; thus, appropriately, there was no Flood Watch issued for these 13 stations.

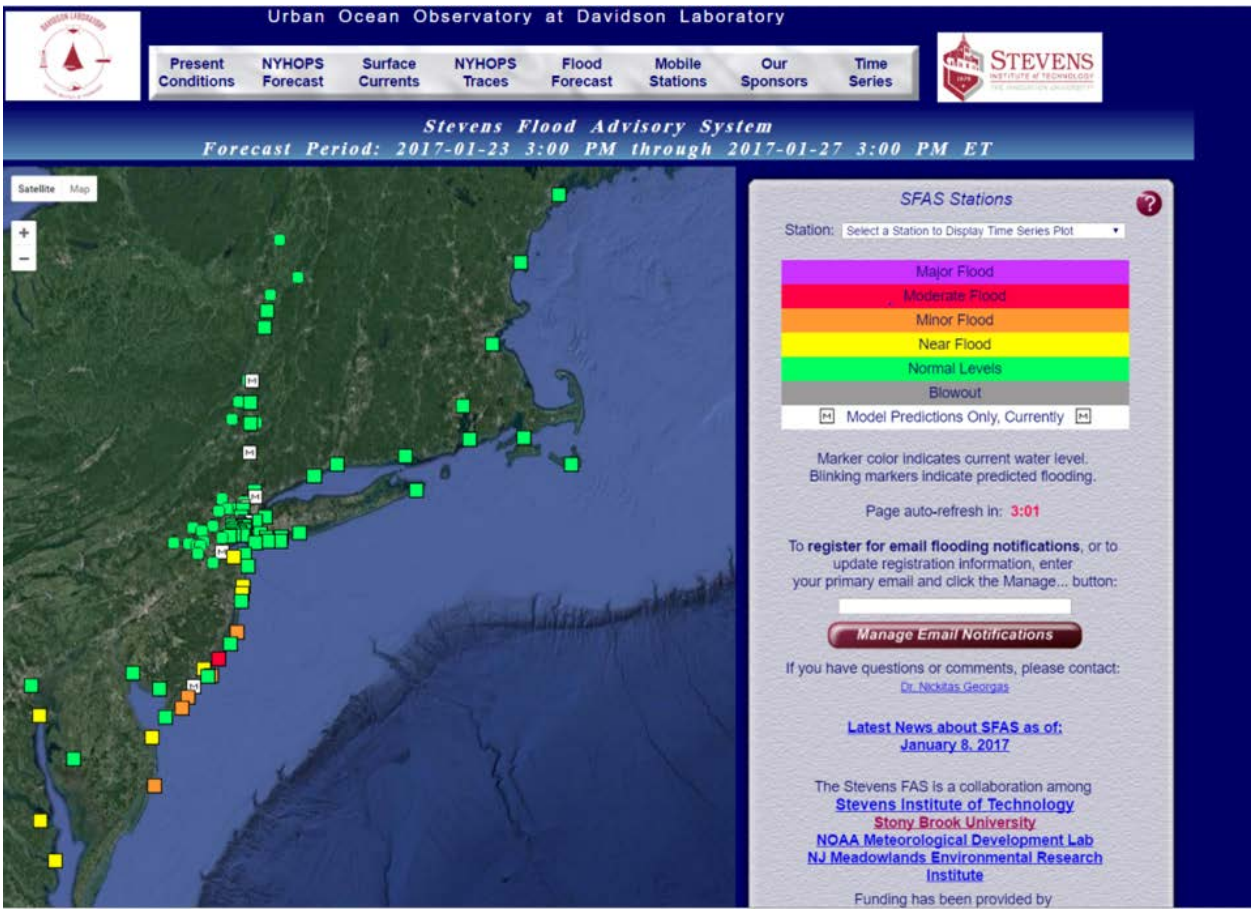
Table 2. Summary of SFAS Flood Advisories for Nor'Easter Kori

Hits: 30 (stations that received both early flood watches and day-of-the-storm advisories, and flooded) **False Alarms:** 0 (received advisory but no flood occurred). **Misses:** 16 (stations flooded, but no advisory was issued; 3 received an early Flood Watch, while 13 were not warned against Minor Flooding of 0.1-0.7ft).

PART 2.

Snapshots of storm surges experienced during Nor'Easter Kori

January 23-24 2017



Sample Stations:

- The Battery NY
- Kings Point NY
- Sandy Hook NJ
- Atlantic City NJ
- New Haven CT
- Boston MA
- Newport RI
- Lewes DE

How to read the graphs of the Stevens Flood Advisory System

In the figure for each station:

- **The graph on top left corner** includes basic information for each station, such as station's name, start date and end date, datum, forecast model displayed, unit and time zone.
- **The graph on bottom left corner** is the map of the location of the selected station.
- **The graph on top right corner shows water level information:**

~Predicted water level based on Stevens Ensemble Forecast models (line)

~Ensemble uncertainty from 5% to 95% (Gray area)

~Observed water level (Red dots, where available)

~Time marker when the snapshot was taken (vertical black dash line)

Note: For Boston MA station, the Predicted water level is based on Stevens SNAP Extended Ensemble Forecast model (dark green line); the Stevens NYHOPS Ensemble is used for the rest of the stations (pink line)

- **The graph on bottom right corner shows storm surge information:**

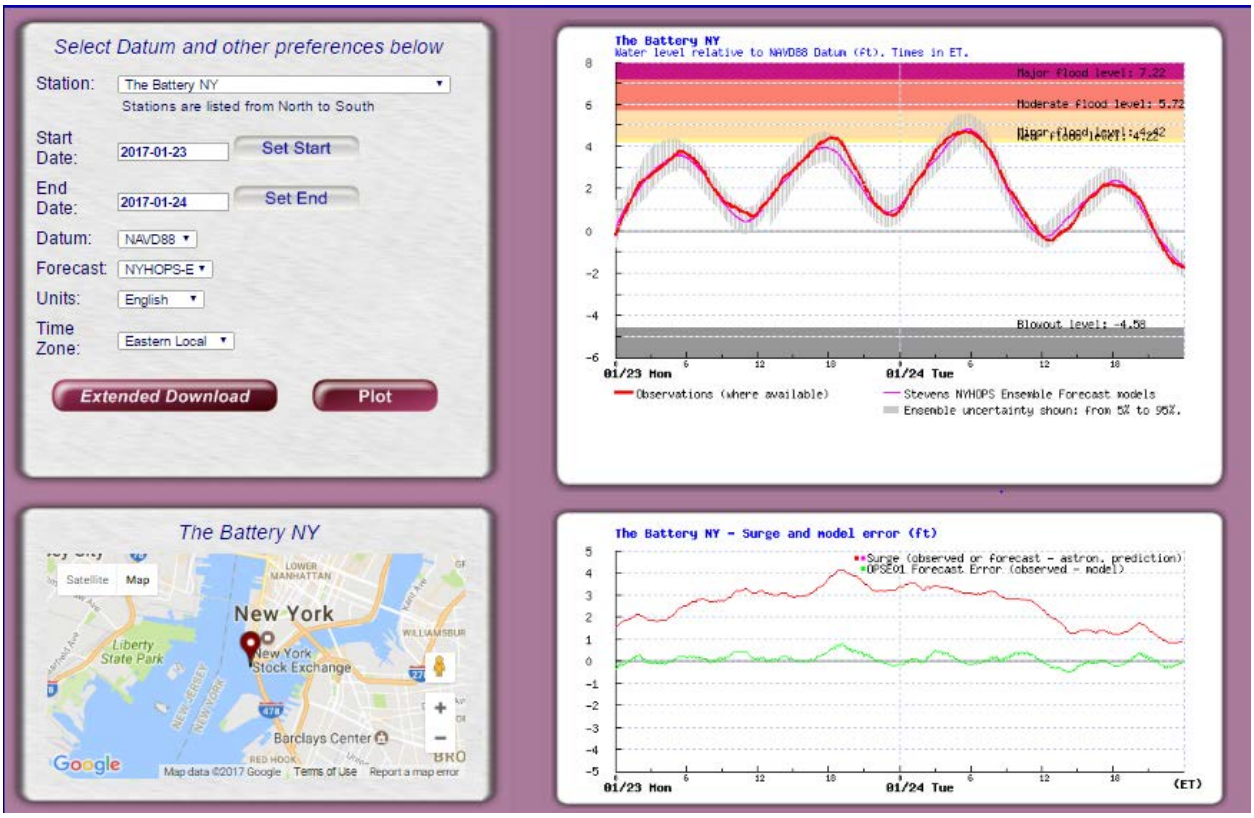
~Observed surge = observed water level – predicted astronomical tide (Red line)

~Forecast surge = predicted water level – predicted astronomical tide (Pink line)

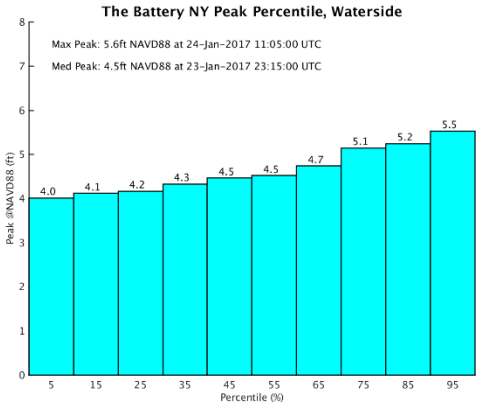
~Forecast error = observed surge – model surge (Green line)

~Time marker when the snapshot was taken (vertical black dash line)

Figures for sample stations:

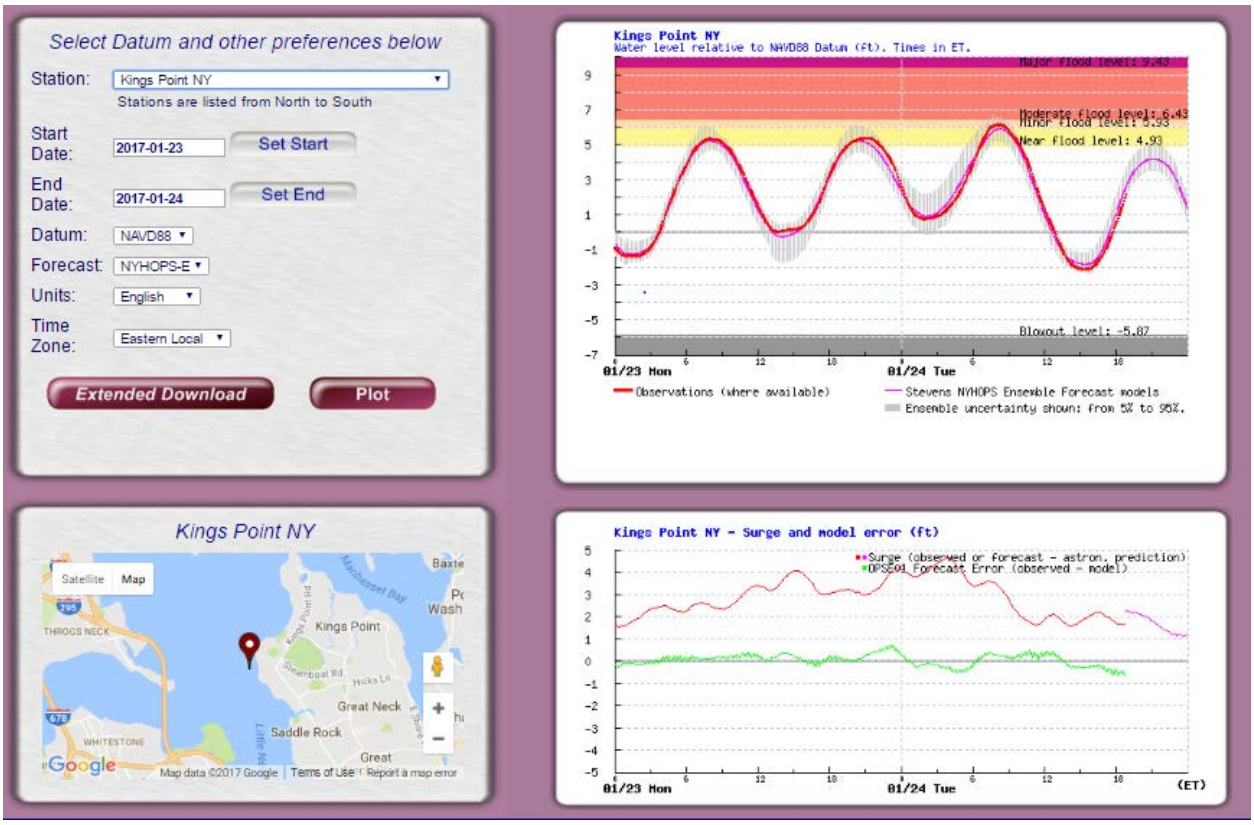


The Battery NY

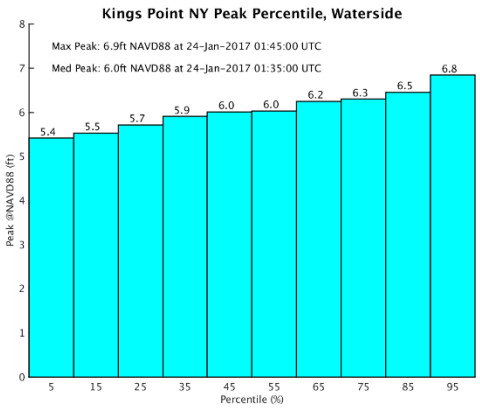


Percentiles of 96hr-prior forecast peaks.

Verified peak was 4.72ft NAVD88 on January 24 2017, 10:30 UTC.

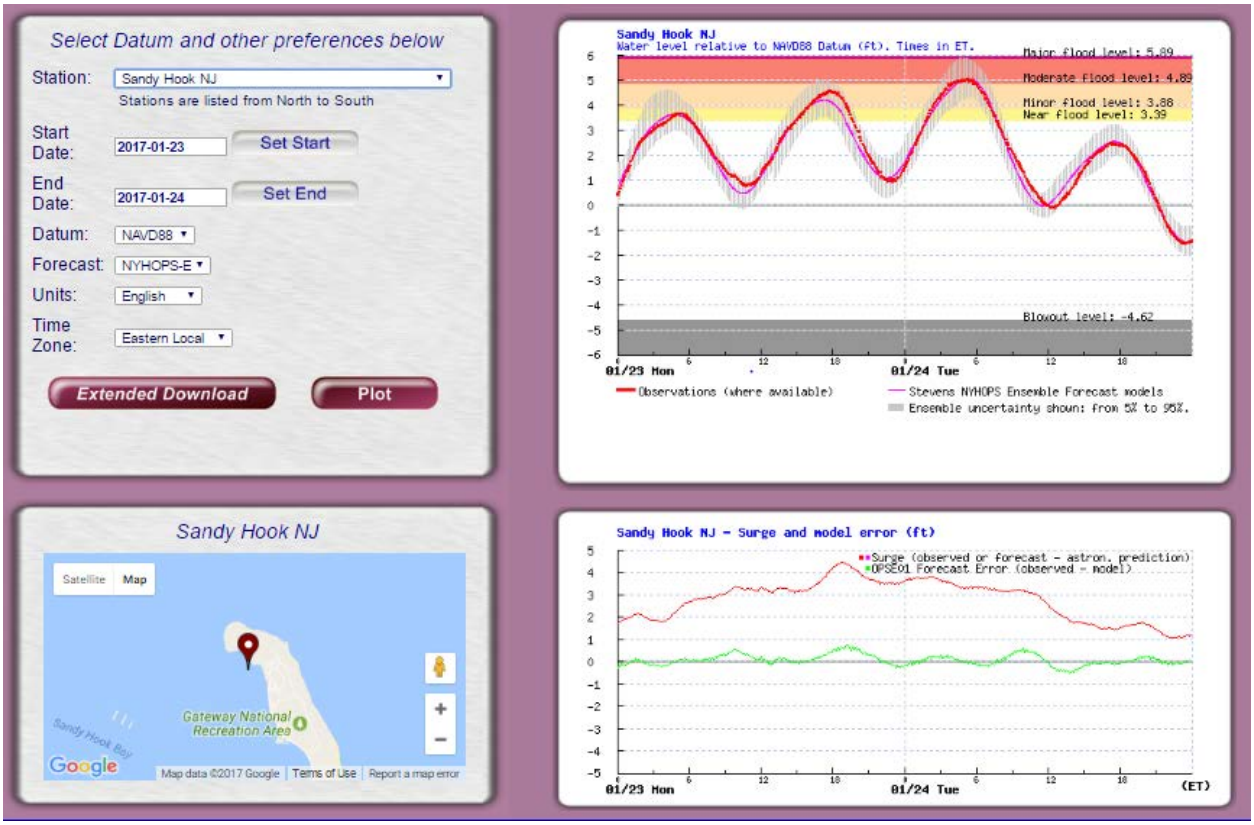


Kings Point NY

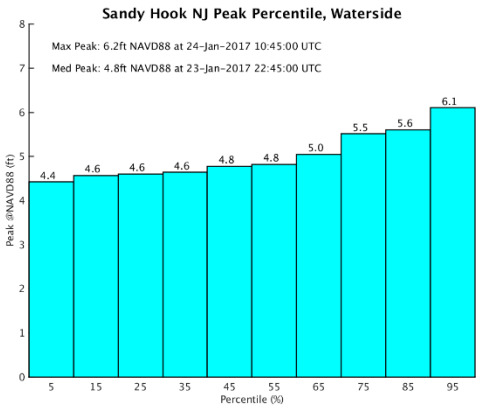


Percentiles of 96hr-prior forecast peaks.

Verified peak was 6.17ft NAVD88 on January 24 2017, 13:00 UTC.

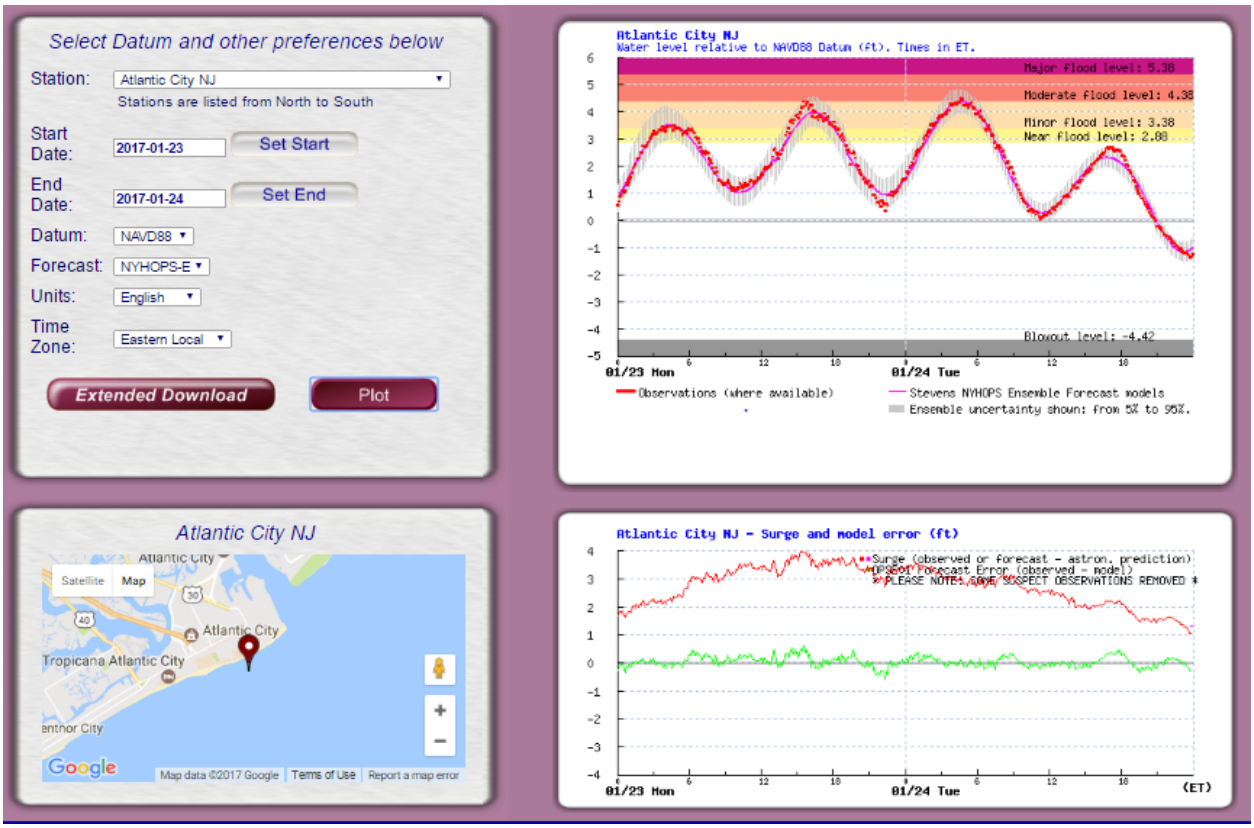


Sandy Hook NJ

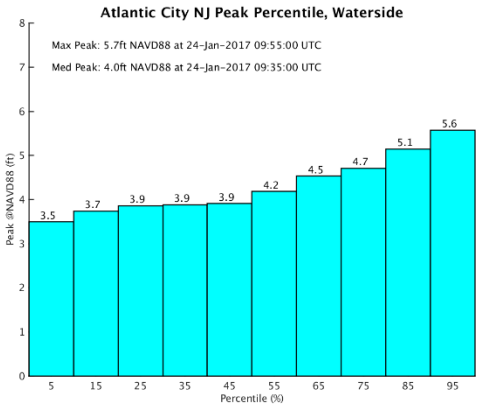


Percentiles of 96hr-prior forecast peaks.

Verified peak was 5.08ft NAVD88 on January 24 2017, 10:12 UTC.



Atlantic City NJ

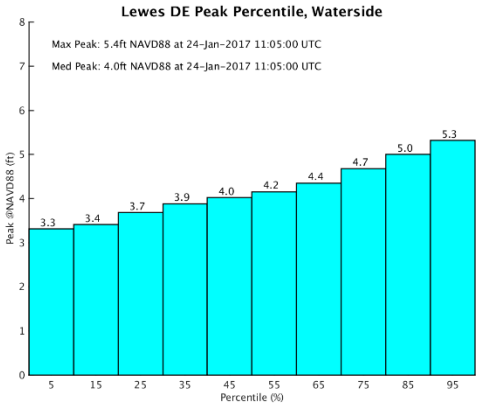


Percentiles of 96hr-prior forecast peaks.

Verified peak was 4.49ft NAVD88 on January 24 2017, 9:36 UTC.

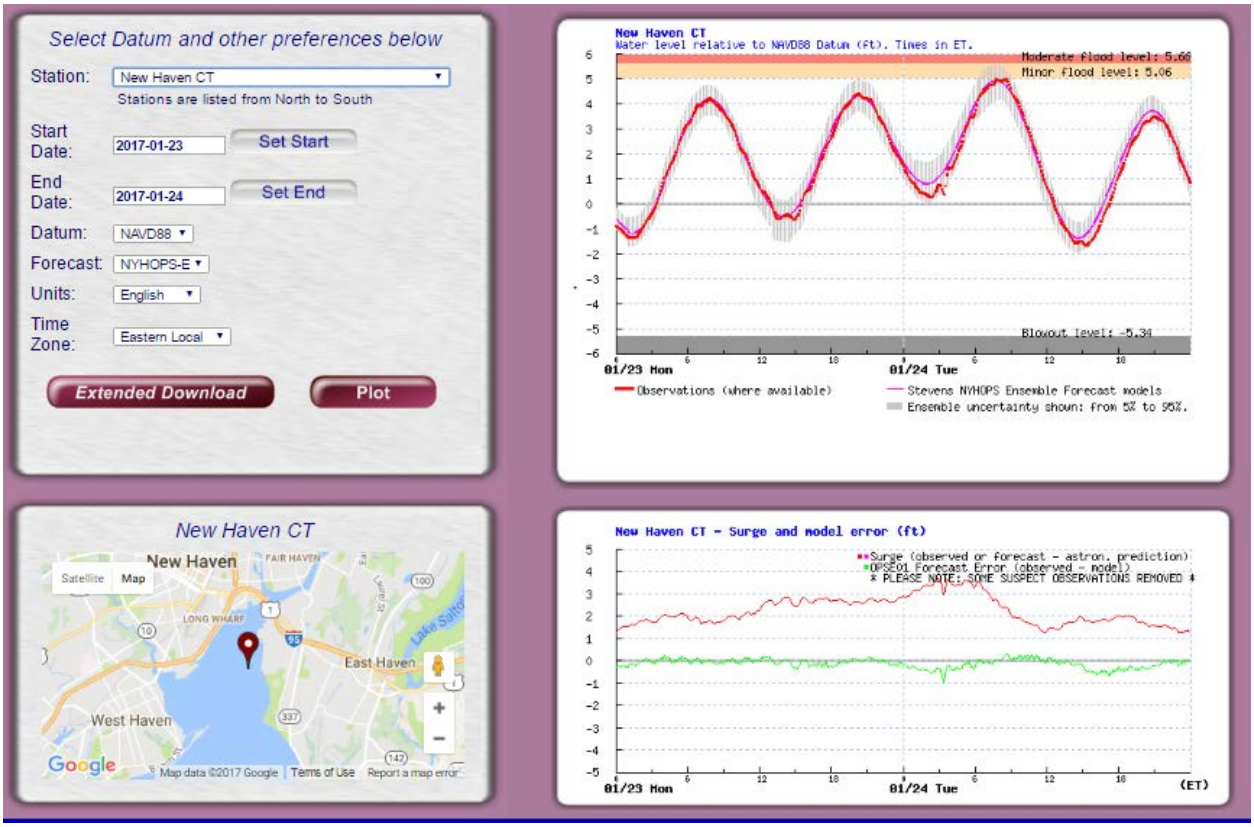


Lewes DE

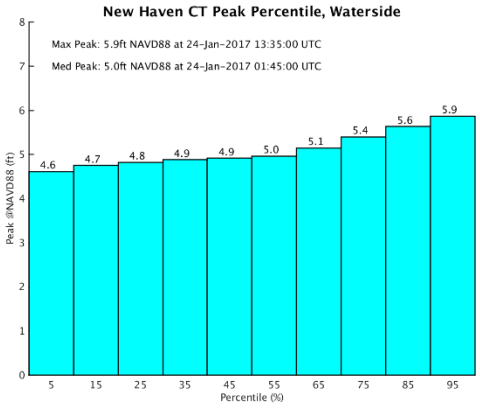


Percentiles of 96hr-prior forecast peaks.

Verified peak was 4.47ft NAVD88 on January 23 2017, 22:12 UTC.

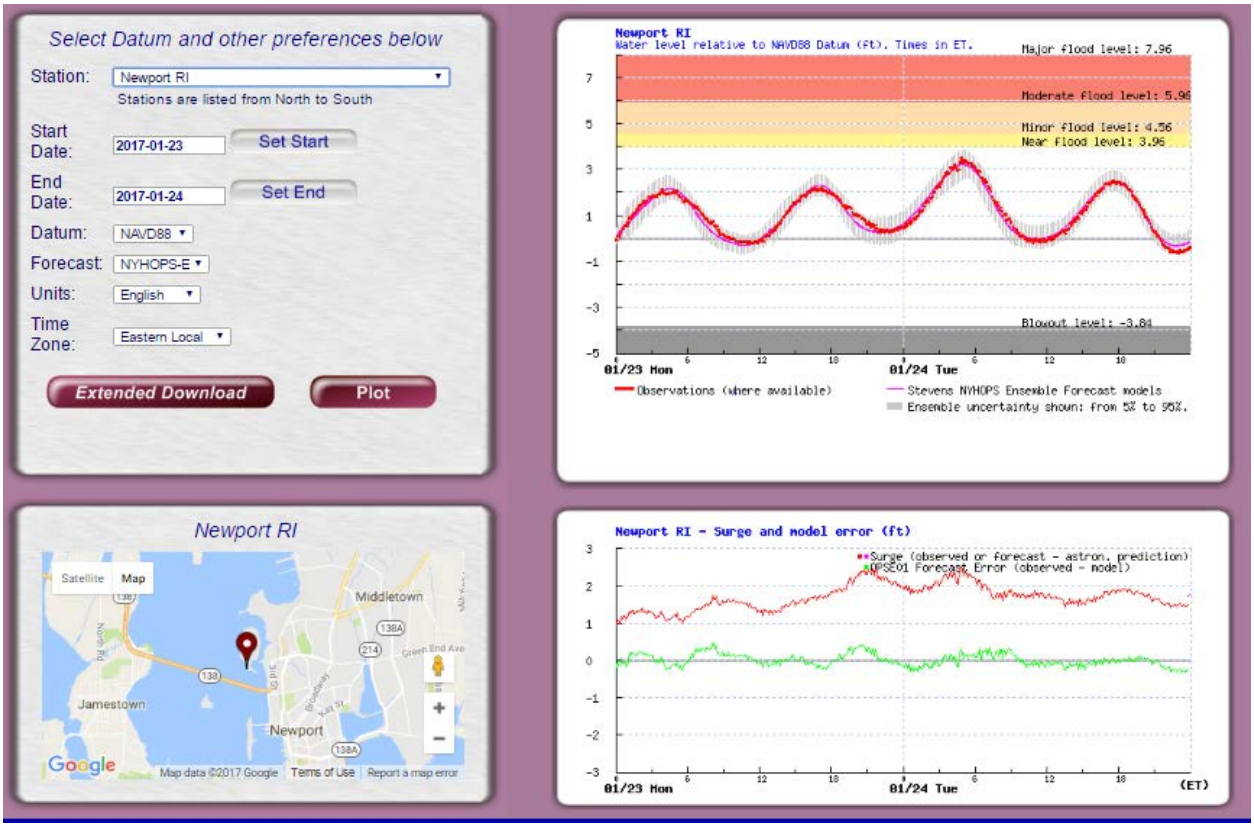


New Haven CT

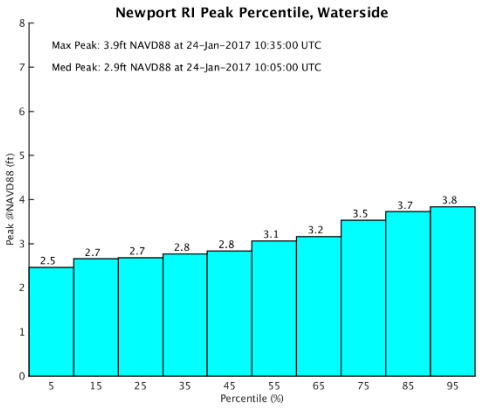


Percentiles of 96hr-prior forecast peaks.

Verified peak was 5.01ft NAVD88 on January 24 2017, 12:48 UTC.

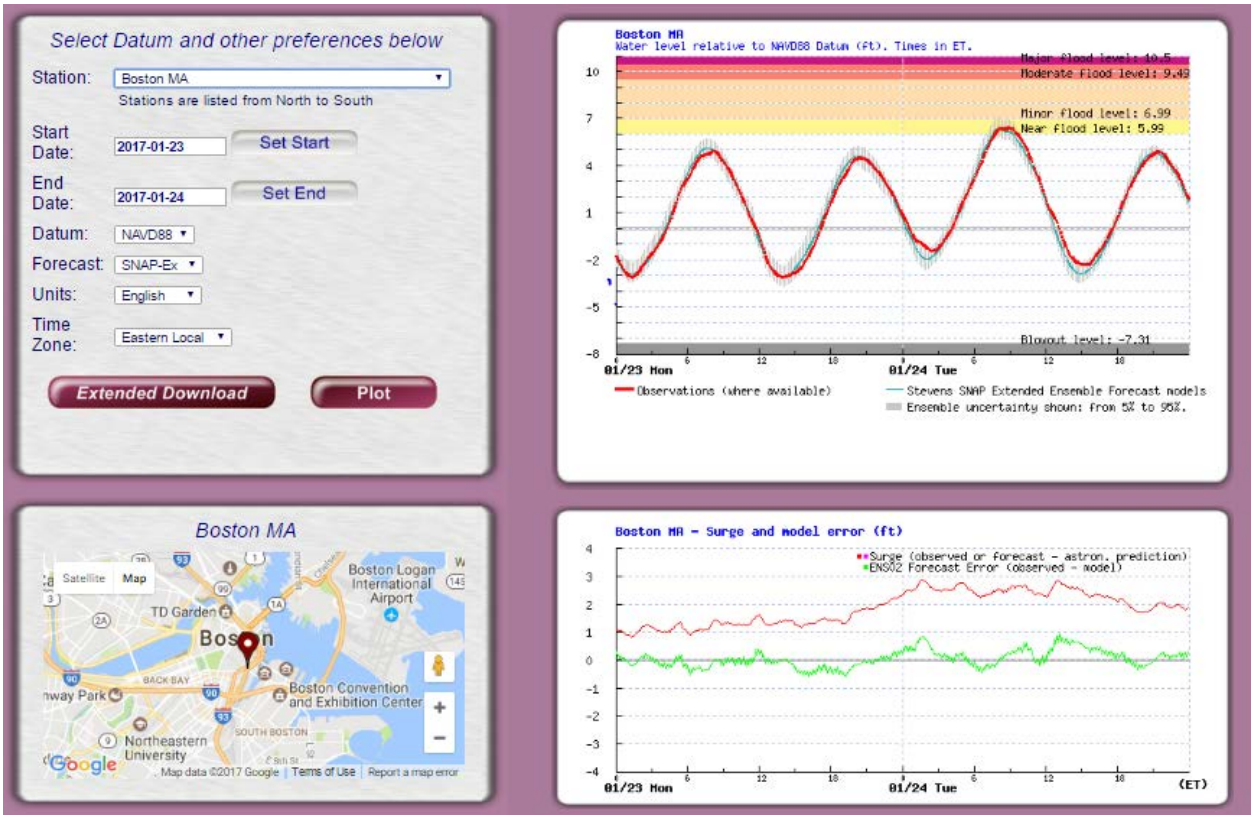


Newport RI



Percentiles of 96hr-prior forecast peaks.

Verified peak was 3.51ft NAVD88 on January 24 2017, 9:48 UTC.



Boston MA

Finally, Table 3 (next page) shows the observed or, in the absence of observations, simulated peaks for the storm.

Table 3. Preliminary Peak Water Levels for Nor'Easter Kori at Stevens Flood Advisory System Stations

Station Name	Peak Water Level (ft NAVD88)	Peak time (EST)	NWS Flood Level
Portland ME	5.73	1/24/2017 8:24	
Fort Point NH	5.89	1/24/2017 8:30	Near
Boston MA	6.45	1/24/2017 8:48	Near
Providence RI	3.61	1/24/2017 5:06	Near
Newport RI	3.51	1/24/2017 4:48	
Woods Hole MA	3.05	1/24/2017 5:48	
Nantucket MA	2.76	1/24/2017 8:54	
New London CT	3.43	1/24/2017 5:42	Minor
New Haven CT	5.01	1/24/2017 7:48	
Bridgeport CT	5.37	1/24/2017 8:00	Minor
Hudson River at Albany NY	5.35	1/24/2017 13:15	
Schodack Island NY	5.15	1/24/2017 13:00	
Norrie Point NY	3.56	1/24/2017 10:15	Near
Hudson River at Piermont NY	4.55	1/24/2017 7:00	Near
Hackensack River at Hackensack NJ	4.93	1/23/2017 19:54	Minor
Barge Marina at Carlstadt NJ	5.1	1/24/2017 7:00	Minor
Passaic River near Harrison NJ	5.05	1/24/2017 5:00	
Passaic River at PVSC at Newark NJ	5.01	1/24/2017 4:48	Minor
Elizabeth Channel near Newark NJ	5.13	1/24/2017 4:36	Near
Newark Bay near Elizabeth NJ	4.84	1/24/2017 4:54	
Arthur Kill near Bloomfield NY	4.84	1/24/2017 4:48	Minor
Bergen Point West Reach NY	4.94	1/24/2017 4:48	Minor
Kearny Point at Kearny NJ	5.03	1/24/2017 6:00	Minor
New York Harbor near Bayonne NJ	4.65	1/24/2017 5:06	
Hudson River near Hoboken NJ	4.56	1/24/2017 5:30	Minor
Hudson River near Hudson Square NY	4.56	1/24/2017 5:36	
The Battery NY	4.72	1/24/2017 5:30	Minor

New York Harbor near Red Hook NY	4.38	1/24/2017 5:36	Minor
New York Harbor near Bay Ridge NY	4.73	1/24/2017 5:18	
Bowery Bay at East Elmhurst NY	5.7	1/24/2017 8:36	Near
Kings Point NY	6.17	1/24/2017 8:00	Minor
Bergen Basin at Jamaica Bay NY	4.62	1/24/2017 4:42	Minor
Jamaica Bay at Inwood NY	4.91	1/24/2017 4:48	Minor
East Rockaway Inlet at Atlantic Beach NY	4.68	1/24/2017 4:42	Moderate
Reynolds Channel at Point Lookout NY	4.6	1/24/2017 5:18	Moderate
Great South Bay at Lindenhurst NY	3.32	1/24/2017 6:42	Major
Montauk NY	3.78	1/24/2017 5:18	Minor
Raritan Bay at Keansburg NJ	5.15	1/24/2017 4:54	Moderate
Sandy Hook NJ	5.08	1/24/2017 5:12	Moderate
Shrewbury River at Sea Bright NJ	5.01	1/24/2017 6:24	Moderate
Shark River at Belmar NJ	5.04	1/24/2017 4:54	Moderate
Watson Creek at Manasquan NJ	4.51	1/24/2017 5:12	Moderate
Barnegat Bay at Mantoloking NJ	2.45	1/24/2017 10:42	Moderate
East Thorofare at Ship Bottom NJ	3.72	1/24/2017 7:00	Moderate
Little Egg Inlet near Tuckerton NJ	4.36	1/24/2017 5:30	Major
Absecon Creek Rte 30 at Absecon NJ	4.74	1/23/2017 17:30	Moderate
Inside Thorofare Rte 40 Atlantic City NJ	4.62	1/24/2017 5:30	Minor
Atlantic City NJ	4.49	1/24/2017 4:36	Moderate
Ludlam Thorofare at Sea Isle City NJ	4.86	1/23/2017 16:54	Moderate
Great Channel at Stone Harbor NJ	4.65	1/23/2017 17:24	Minor
Cape May NJ	4.34	1/24/2017 5:24	Minor
Maurice River at Bivalve NJ	4.26	1/24/2017 6:30	Minor
Lewes DE	4.47	1/23/2017 17:12	Moderate
Ocean City Inlet MD	2.99	1/23/2017 16:36	Minor
Cambridge MD	2.31	1/24/2017 0:54	Near
Annapolis MD	1.81	1/24/2017 1:06	Minor
Baltimore MD	1.64	1/24/2017 3:30	

Lewisetta VA	2.43	1/23/2017 22:18	Minor
Windmill Point VA	2.19	1/23/2017 20:36	Minor
Barnegat Bay at Barnegat Light NJ	3.37	1/24/2017 4:30	Minor
Chesapeake Bay Bridge Tunnel VA	3.27	1/24/2017 5:36	Minor
Duck NC	3.39	1/24/2017 4:30	Minor
Tivoli Bays South NY	5.05	1/24/2017 10:55	Minor
Hudson River below Poughkeepsie NY	4.19	1/24/2017 9:30	Minor
Hudson River at South Dock at West Point NY	4.32	1/24/2017 8:35	Minor
Hudson Rv South of Hastings-on-Hudson NY	4.6	1/24/2017 6:55	Minor
FDU Bridge at Hackensack NJ	5.67	1/24/2017 7:25	Minor
Saw Mill Creek at Kearny NJ	5.01	1/24/2017 6:15	Minor
Hudson River near Jersey City NJ	4.47	1/24/2017 5:55	Minor
Hudson River at 39th St NY	4.4	1/24/2017 6:05	
Raritan Bay at South Amboy NJ	5.34	1/24/2017 5:25	Moderate
Rockaway Inlet near Floyd Bennett Field NY	5	1/24/2017 5:06	Minor
Hudson Bay at Freeport NY	4.51	1/24/2017 5:18	Moderate
Peck Bay at Ocean City NJ	5.28	1/24/2017 5:05	Moderate
Cohancey River at Greenwich NJ	3.85	1/23/2017 8:00	Minor